

Table 6: Total spectral energy across precision modes
Muon+GRPO BF16 retains only 0.4% of the FP32 spectral energy;
AdamW+GRPO BF16 retains 3.7% — its concentrated top SVs survive rounding better

Condition	Prec.	$\Sigma \sigma_i^2$ (energy)	$\ \Delta W\ _F$	$\Sigma \sigma_i$ (nuclear)	σ_1	median(σ_i)	BF16/FP32 energy %
AdamW+GRPO	FP32	1.47e-03	0.0327	0.6399	1.44e-02	3.06e-04	—
AdamW+GRPO	BF16	5.43e-05	0.0063	0.2156	4.87e-04	1.55e-04	3.7%
AdamW+GRPO	Mixed	1.52e-03	0.0334	0.8069	1.23e-02	4.26e-04	103.2%
Muon+GRPO	FP32	6.65e-04	0.0214	0.7469	1.12e-03	5.69e-04	—
Muon+GRPO	BF16	2.75e-06	0.0014	0.0435	1.95e-04	3.10e-05	0.4%
Muon+GRPO	Mixed	6.32e-04	0.0209	0.7271	1.19e-03	5.51e-04	95.0%
AdamW+SFT	FP32	2.05e-03	0.0387	1.0558	1.16e-02	6.16e-04	—
AdamW+SFT	BF16	4.88e-04	0.0187	0.6367	1.67e-03	4.56e-04	23.8%
AdamW+SFT	Mixed	2.99e-03	0.0467	1.1533	1.67e-02	6.47e-04	146.1%
Muon+SFT	FP32	1.58e-03	0.0333	1.1756	1.81e-03	8.95e-04	—
Muon+SFT	BF16	5.08e-06	0.0019	0.0543	3.73e-04	3.70e-05	0.3%
Muon+SFT	Mixed	1.47e-03	0.0321	1.0981	2.05e-03	8.15e-04	93.0%